MICHAEL P. RAGSDALE

Senior Front-End Developer & Accessibility Engineer Vanilla JS (ES6+) • Section 508 Compliance • SPA Architecture Norfolk, VA (Remote) | U.S. Citizen | hireme@michaelpragsdale.com

Professional Summary

Senior Front-End Developer and Systems Architect with over 15 years of experience specializing in custom SPA web platforms and accessibility-first architecture. Creator of the Elara Front Controller, a zero-dependency Vanilla JavaScript (ES6+) SPA routing engine. Deeply experienced in translating complex UI/UX requirements into mobile-first, responsive applications. Proven track record in strict government and academic environments, specializing in massive-scale Section 508 and WCAG 2.1 AA compliance remediation and governance.

Technical Skills

- **Front-End & UI:** JavaScript (ES6+), HTML5, CSS3, SPA Architecture, DOM Manipulation
- **Accessibility:** WCAG 2.1 AA, Section 508, U.S. Web Design System (USWDS) principles, Deque axe
- **Infrastructure & Tooling:** Git, GitHub/GitLab Workflows, RESTful APIs, Webpack/Build Pipelines
- **Design Integration:** Adobe Creative Suite, Mockup Translation, Component-Based Architecture

Relevant Technical Experience

Principal Systems Architect & Founder

RaggieSoft Media | Norfolk, VA | raggiesoftmedia.com | Est. 2008 – Present Founded and operate RaggieSoft Media, an independent software development and digital media publishing ecosystem supporting web applications, cloud infrastructure, narrative publishing, music distribution, and open-source platform development.

- **Platform Architecture & CMS:** Architected and open-sourced **Stardust Engine CMS**, an MIT-licensed, zero-database PHP MVC platform utilizing a lightweight JSON data layer, native PHP request routing, and **Elara**, a custom Vanilla JavaScript SPA routing engine for dynamic DOM hydration.
- **Multi-Domain & Cloud Edge:** Orchestrate a production fleet of 9 distinct domains via **Cloudflare DNS/CDN** edge routing, 301 redirection rules, and automated deployment pipelines syncing assets to **DigitalOcean Spaces** (S3-compatible object storage) and Nginx reverse proxies.
- **Automated Media Engineering:** Engineered automated server-side pipelines utilizing **FFmpeg** and Bash to transcode and normalize high-resolution audio assets across multiple distribution formats (FLAC, WAV, MP3, OGG), paired with custom **PHP GD** engines that dynamically composite Open Graph social cards across routes.
- **E-Commerce & Digital Fulfillment:** Integrated **Fourthwall** as an international Merchant of Record (MoR) with custom DNS architecture, powering automated, frictionless global fulfillment of physical merchandise and digital assets without third-party vendor lock-in.
- **AI Governance & Engineering:** Authored comprehensive architectural constraints, repository standards, and prompt protocols to govern AI-assisted engineering workflows across the codebase, ensuring maintainability, code integrity, and architectural consistency.
- **Accessibility by Design:** Enforce strict **WCAG 2.1 AA** and **Section 508** compliance across all UI components, custom modals, media players, and narrative readers, ensuring zero accessibility debt out of the box.

Data Normalization & Technical Lead

City of Virginia Beach Parks & Recreation | Virginia Beach, VA | May 2023 – Present

- **Compliance Enforcement:** Govern output data to guarantee strict Section 508 compliance across all generated operational documents from legacy systems, ensuring institutional adherence to federal

accessibility mandates.
- **ETL Pipelines:** Architect and manage manual ETL pipelines to ingest and normalize complex operational data into highly accessible, mobile-friendly digital formats.

Section 508 Compliance Specialist (Computer Operations)

Old Dominion University (ODU Online) | Norfolk, VA | Nov 2019 – Jul 2022

- **Enterprise Accessibility Strategy:** Served as the primary technical gatekeeper for enterprise-level web content during a massive, university-wide digital infrastructure migration to the Canvas LMS.
- **Compliance Auditing:** Executed rigorous WCAG 2.1 AA and Section 508 compliance audits, remediating complex codebases (HTML/CSS) and ensuring strict structural integrity across institutional platforms using tools like Deque axe.
- **Architectural Standardization:** Engineered universal, inherently accessible architectural templates deployed across the university ecosystem, heavily streamlining workflows for instructional design faculty.

Mobile Applications Developer Intern

Dominion Enterprises | Norfolk, VA | Sep 2012 – Dec 2012

- Engineered cross-platform mobile applications for iOS and Android utilizing the JavaScript-based Titanium SDK.
- Gained foundational, hands-on experience in component-based mobile architecture and deployment.

Education & Certification

- **Tidewater Community College | Virginia Beach, VA** Associate of Science, Information Technology (Focus: Software Engineering & Web Dev) | Expected Dec 2027
- **Old Dominion University | Norfolk, VA** Bachelor of Science, Customized Studies in Leadership | Prior Coursework
- **Wilson Workforce and Rehabilitation Center (formerly Woodrow Wilson Rehabilitation Center) | Fishersville, VA** Certificate, Business & Information Technology | 2011